UPDATES

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# Preparation and Evaluation of Sterically Hindered Acridine Photocatalysts

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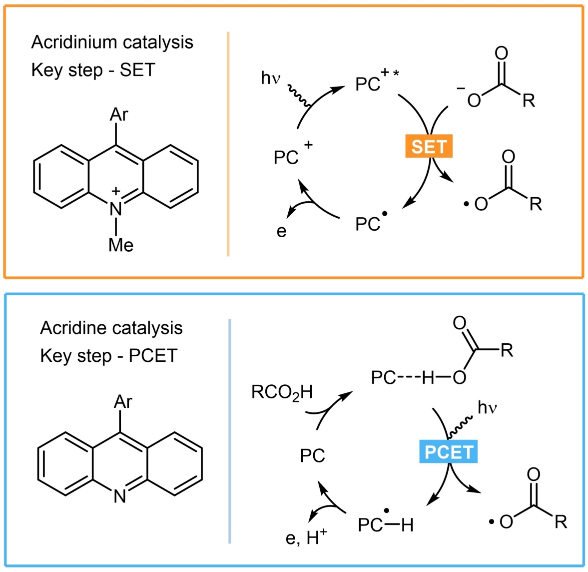
**Abstract:** A practical nickel-catalyzed protocol for the synthesis of sterically hindered 9-arylacridines via Negishi type cross-coupling was developed. The method enables a convenient approach to a set of active acridine photocatalysts starting from commercially available 9-chloroacridine, inexpensive catalyst $[NiCl_2(Ph_3P)_2]$, and organozinc reagent with no need for additional ligands and precious metals. The synthetic protocol could be readily performed on a gram scale. The photocatalytic efficiency of obtained acridines was tested in decarboxylative thiolation and amination reactions. Structures of 19 acridines were studied by X-ray analysis, which allowed to propose a steric descriptor relevant to the catalytic activity.

**Keywords:** Acridine; Photocatalysis; Kumada cross-coupling; Nickel




**Scheme 1.** Acridine-based catalysis.

## Introduction

Acridine core has long been recognized as a valuable heterocyclic framework[1] for versatile applications in drug design[2] and material sciences.[3] With the advent of photocatalysis,[4] the ability of acridine system to absorb visible light was harnessed to promote photoinduced organic transformations. Two types of light-promoted catalysis based on 9-aryl-substituted acridine derivatives have been realized (Scheme 1). First, *N*-acridinium salts were used as metal-free photooxidants,[5] and were applied towards electron-rich alkenes, aromatics, and carboxylate anions.[6,7] Acridinium salts typically follow the regular photoredox catalytic pathway serving as SET oxidants via reductive quenching.[6] Another mechanistic option involves acridines themselves, which upon light irradiation can activate acidic species via the proton-coupled electron transfer (PCET) mechanism.[8] Following the pioneering observations by Okada,[9] this concept was used by Larionov[10] and recently by our group[11] to activate free carboxylic acids. We also applied acridine-type catalysis for the activation of the S–H bond.[12]

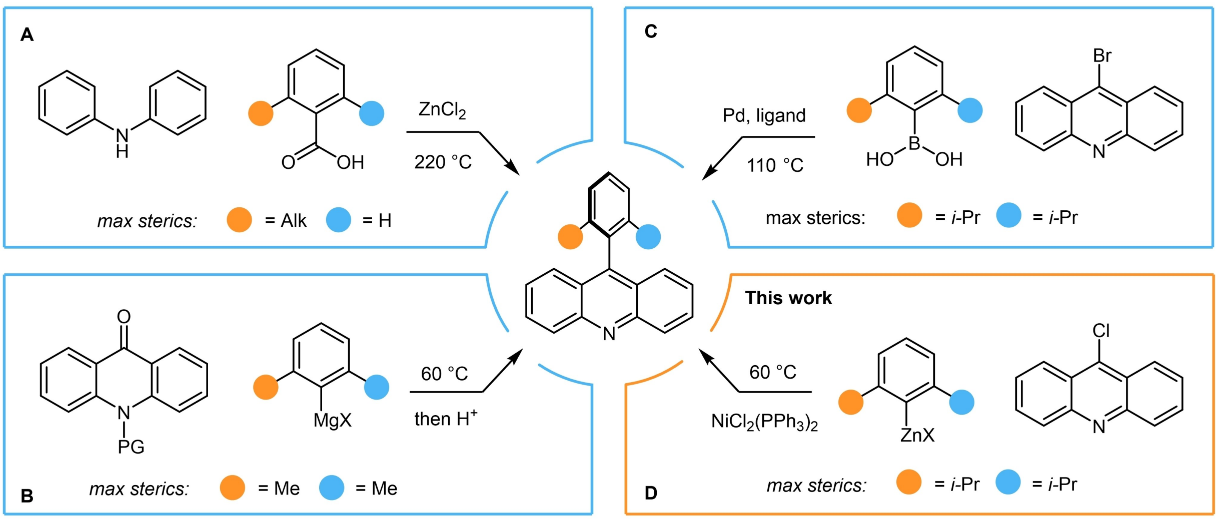
Acridines have a significant advantage over iridium-based and acridinium oxidants towards primary carboxylic acids. Indeed, as was shown by Larionov,[10] acids leading to primary alkyl radicals were efficiently activated by acridines, while iridium photocatalysts and acridinium salts are notably less effective with these substrates.[13]

The major drawbacks of acridine photocatalysis include: (a) high loadings of weighty acridines (5–15 mol%), and the reason for this has not been clarified; and (b) laborious methods for the synthesis of some acridines. During the course of our research on acridine-driven photochemistry,[11,12] it was noted that acridines bearing *ortho*-substituted aromatic group in the C-9 position generally provided more satisfying results.

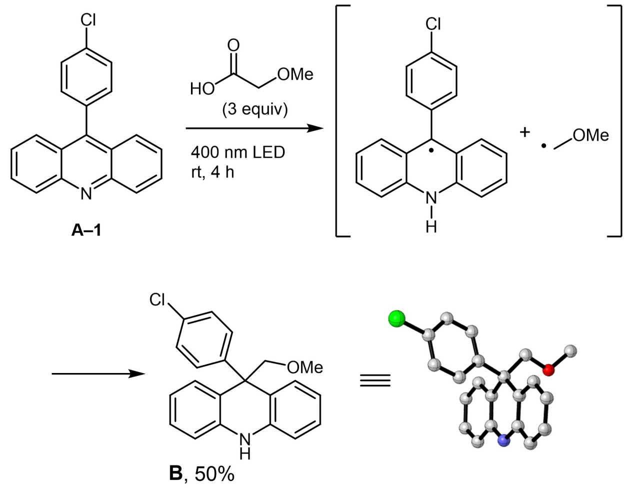
Herein we provide a rationale for the degradation of the catalyst, which explains the need for *ortho* substituents in the aryl group. We also report on a convenient protocol for the synthesis of sterically hindered acridines. Concerning the synthesis of 9-aryl-substituted acridines, despite various approaches that have been developed, general and simple methods for sterically hindered compounds are still lacking. Thus, Bernthsen-type reactions represent the most common approach to acridine core construction, but it requires extreme heating under acidic conditions.[14] (Scheme 2A). Following a similar logic, efficient and mild protocols were developed using *ortho*-phenylamino benzonitriles.[15] or *ortho*-aminophenyl benzophenones.[16] as starting materials. Besides apparent functional group limitations, these reactions cannot be applied to the synthesis of bulky acridines (e.g., Ar=2,6-diisopropylphenyl). The addition of Grignard reagents to *N*-protected acridone followed by aromatization was used for the preparation of a 9-mesitylacridine catalyst (Scheme 2B).[10f,17] However, we failed to obtain acceptable results for the introduction of a bulkier aryl group. Several palladium-catalyzed cross-coupling protocols involving boronic acids were developed (Scheme 2C).[18] However, the only reported method to access 9-(2,4,6-triisopropyl) acridine requires a sophisticated oxaphosphole ligand, while it was emphasized by the authors that ligands from a commercial toolkit (e.g. X-Phos, Ru-Phos, S-Phos, dppf) gave only trace amounts of coupling product in this process.[18c] Herein we demonstrate that various 2,6-disubstituted aryls can be conveniently introduced via nickel-catalyzed Negishi cross-coupling[19] (Scheme 2D).

## Results and Discussions

*Catalyst deactivation study.* To elucidate the fate of the photocatalyst, an acridine **A-1**, which does not have substituents at the *ortho*-positions of the 9-aryl group, was combined with an excess amount of a carboxylic acid under typical conditions but without additional radical trapping agents. Thus, 9-(4-chlorophenyl)acridine (**A-1**) was irradiated in the presence of an excess of methoxyacetic acid with 400 nm light in methylene chloride at room temperature. (Scheme 3). Fast acridine consumption was observed, affording product **B**, which was isolated in 50% yield, and its structure was firmly established by X-ray analysis. A similar experiment using 9-mesitylacridine gave a complex mixture. The formation of compound **B** likely proceeds via recombination of acridine-derived radical species with the alkyl radical generated after PCET and decarboxylation, respectively. Considering the typical catalytic reactions, such a way of acridine catalyst degradation may become pronounced if the rate of interaction of alkyl radical with another reacting component (e.g. alkene) is low. It is also apparent that putting additional groups at the *ortho*-positions of the 9-aryl group of the acridine would decrease this acridine deactivation pathway,




**Scheme 2.** Synthesis of sterically hindered acridines.

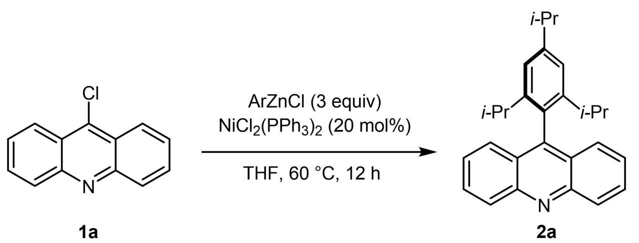



**Scheme 3.** Catalyst degradation experiment.

thereby increasing the catalyst performance. Recently, the influence of steric and electronic effects of the C-9 substituents on catalytic performance of an acridine was studied by Larionov.[10f]

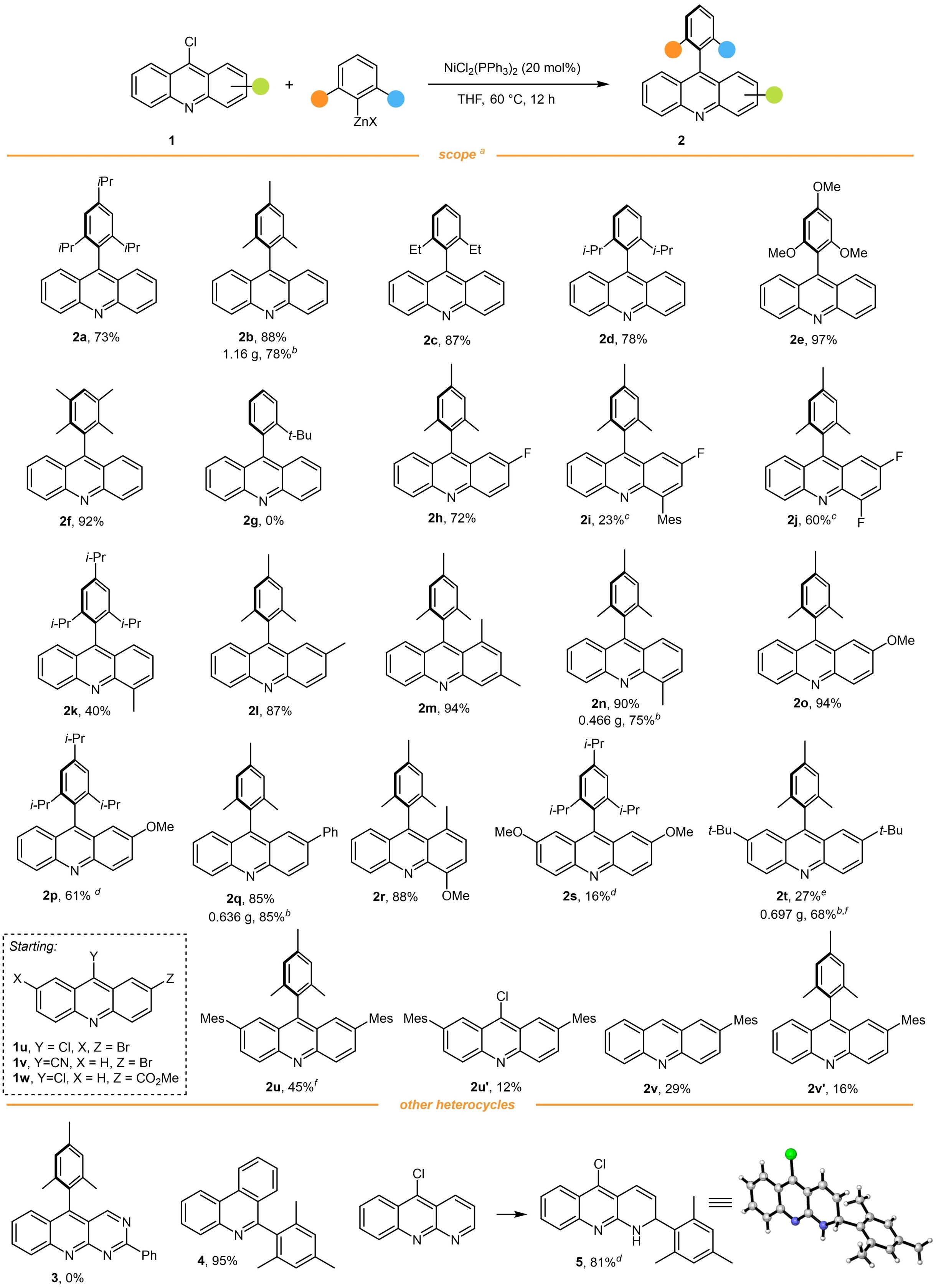
*Cross-coupling reaction.* 2,4,6-Triisopropylphenyl zinc chloride, which was prepared from the corresponding Grignard reagent and zinc dichloride, and 9-chloroacridine **1 a** were selected as model substrates, and their coupling was evaluated (Table 1). The reaction proceeded smoothly in THF solution at 60 °C in the presence of nickel dichloride triphenylphosphine complex $[NiCl_2(Ph_3P)_2]$ (20 mol%) affording **2 a** in 73% isolated yield (entry 1). The use of the parent Grignard reagent led to a slightly decreased yield (entry 2) but gave poor results for most of the other acridine substrates described below (Supporting Information, Table S2). The corresponding boronic acid was completely ineffective under these reaction conditions (entry 3). The nickel complexes with a bidentate phosphine or dimethoxyethane (dme) were ineffective (entries 5 and 6). Noteworthy, analogous palladium catalyst failed to deliver even a trace amount of the coupling product (entry 7). The heating played a significant role in the cross-coupling. Thus, performing the reaction at 40 °C resulted in decreased yield, while at room temperature the reaction did not proceed (entries 8 and 9). Decreased amount of nickel catalyst (5 mol%) and organometallic reagent (1.5 equiv.) did not lead to yield loss (entry 4), but for further experiments, we decided to use the excess of the reagents to ensure generality, high robustness, and hydrolytic stability of the system (large scale experiment was conducted under economic conditions). Under the optimized conditions, the scope of the coupling reaction was investigated (Scheme 4). Various di-*ortho*-substituted aryl groups (2,4,6-triisopropyl-Ph **2 a**; mesityl **2 b**; 2,6-diethyl-Ph **2 c**; 2,6-diisopropylphenyl **2 d**; 2,4,6-trimethoxy-Ph **2 e**; duryl **2 f**) were installed at the 9-th position of acridine with excellent yields. Reactions were typically performed on 0.5 mmol scale. Acridines (**2 b**, **2 n**, **2 q**, **2 t**) were prepared on a larger scale starting from 2–5 mmol of acridine **1 a** using 5% of nickel catalyst and 1.5 equivalents of the zinc reagent, with the product being isolated by recrystallization instead of chromatography. Unfortunately, we failed to introduce *ortho*-*tert*-butyl-substituted phenyl fragment, as compound **2 g** was not formed. 2-Fluoro-substituted acridine substrate suffered slight fluorine loss under reaction conditions (according to GC-MS analysis), yet the desired product **2 h** was isolated by recrystallization in 72% yield. At the same time, 4-fluorine displayed significant activity towards substitution, and two acridine products (**2 i**, **2 j**) were isolated from a single run utilizing 9-chloro-2,4-difluoroacridine as starting material. The method was tolerant to methyl (**2 k–n**), methoxy (**2 o**, **2 p**), and phenyl (**2 q**) substitution in acridine core, even when the substituents were close to the reaction centre (**2 m**, **2 r**). Dimethoxy-substituted product (**2 s**) was obtained in low yield with a more active magnesium reagent. The decreased yield of **2 s** was probably related to the electron-donating effect of the methoxy groups decelerating oxidative addition step. The formation of the sterically hindered product **2 t** did not occur under standard conditions, but **2 t** was obtained in 27% yield by heating at 100 °C in diglyme. With the increase of reaction time, the yield of acridine **2 t** was improved to 68% for the scaled-up run. To elucidate functional

**Table 1.** Optimization studies.




| Entry | Deviation | Yield of **2 a**, %[a] |
|---|---|---|
| 1 | none | 80 (73)[b] |
| 2 | ArMgBr | 64[b] |
| 3 | $ArB(OH)_2$ | 0 |
| 4 | 1.5 equiv. of ArZnCl, 5% of $NiCl_2(Ph_3P)_2$ | 78 |
| 5 | 5% of $NiCl_2$(dppe) | 5 |
| 6 | 20% of $NiCl_2$(dme) | 24 |
| 7 | 5% of $PdCl_2(PPh_3)_2$ | 0 |
| 8 | 40 °C | 52 |
| 9 | 25 °C | 0 |

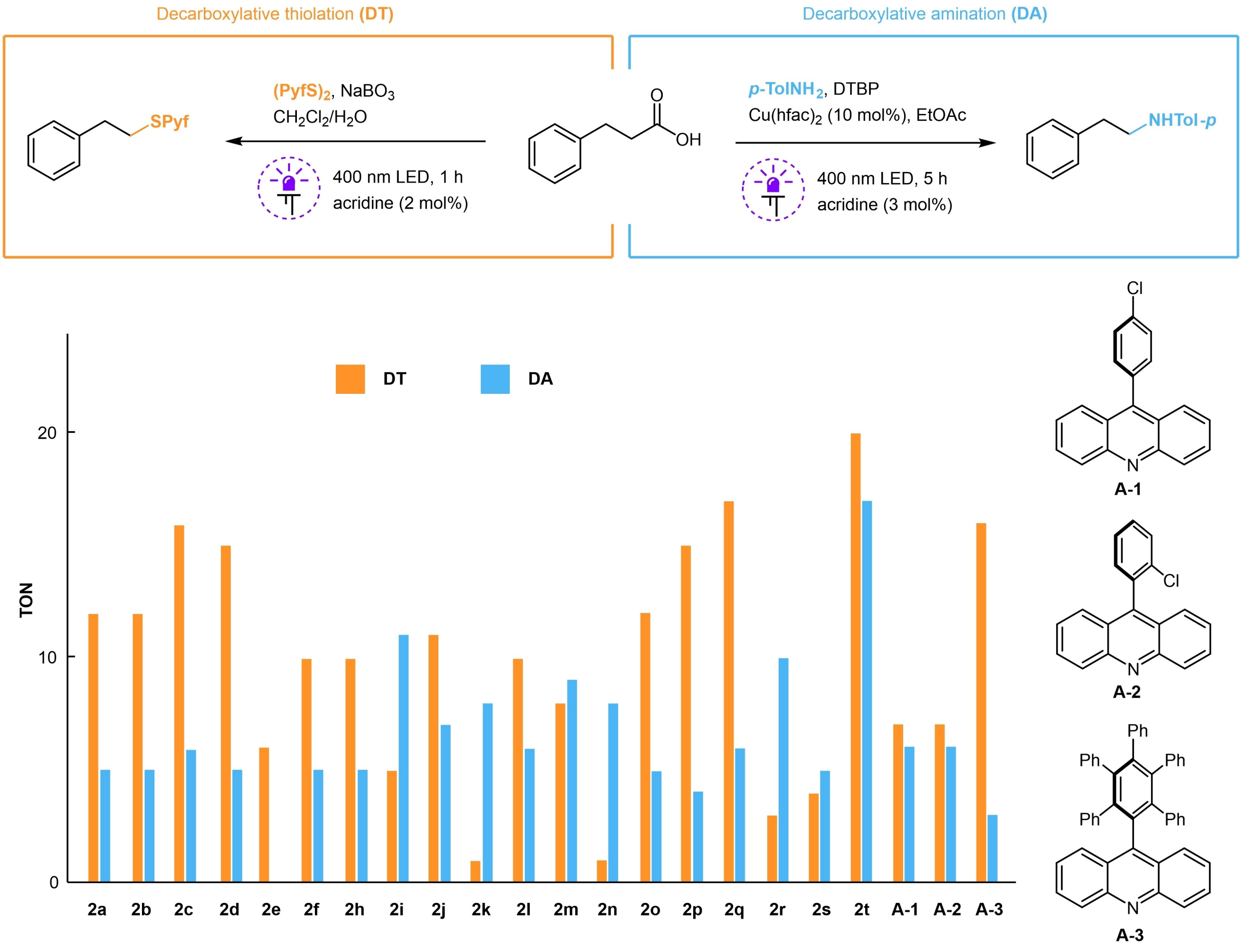
[a] Yield was determined by $^1$H NMR using $CH_2Br_2$ as an internal standard.
[b] Isolated yield.



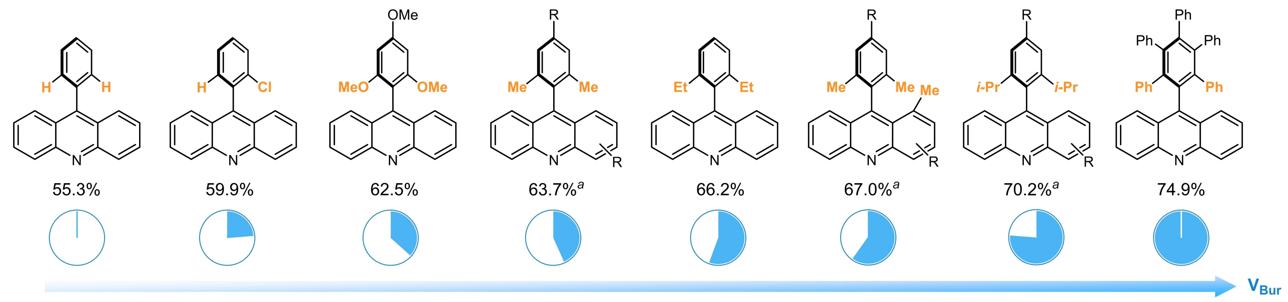

**Scheme 4.** Reaction scope. [a] Isolated yields are shown. [b] 5 mol% of [$NiCl_2(Ph_3P)_2$] and 1.5 equiv. of MesZnCl were used. [c] Products **2i** and **2j** were obtained in a single run starting from 9-chloro-2,4-difluoroacridine. [d] Arylmagnesium reagent was used. [e] Diglyme as solvent, 100 °C, 5 h. [f] Diglyme as solvent, 100 °C, 12 h.

group tolerance, we applied our protocol to 2,7-dibromo-9-chloroacridine **1 u**. Under standard conditions it gave a mixture of mesityl-derived acridines. With the higher reaction temperature, trimesityl acridine **2 u** along with dimesityl acridine **2 u'** were isolated in low yields. The cyano-group in acridine **1 v** suffered protodecyanation and substitution under the action of the nickel catalyst, giving the mixture of **2 v** and **2 v'**. Noteworthy, the ester group of **1 w** was also unstable, leading to an unidentifiable product mixture. Finally, we tested the applicability of the protocol to other related heterocycles. Unluckily, 5-chloro-2-phenylpyrimido[4,5-b] quinoline gave no product (**3**); 6-chlorophenanthridine, on the contrary, delivered a high yield of the coupling product (**4**). Noteworthy, phenanthridine **4** was completely inactive as a photocatalyst in model reactions. The presence of additional nitrogen atom in the conjugated system led to complex mixtures when using mesityl zinc chloride, while with mesityl magnesium bromide the unexpected addition product **5** was formed. The residual nickel content in product **2 b** was found to be 17.5 ppm (ICP-OES).

Further, we evaluated the efficiency of obtained acridines as photocatalysts. For this purpose, we selected two reactions: decarboxylative thiolation (DT) recently reported by our group[11b] (Scheme 5, left) and copper-catalyzed decarboxylative amination (DA) introduced by Larionov[10e] (Scheme 5, right). Turn over number (TON)[20] was utilized as a key performance parameter. Known acridines **A-1**, **A-2**, and **A-3** were included in this study. Decreased loads of catalysts were used, and the degradation of acridine photocatalyst was controlled by GC-MS. The product yields were determined by calibrated GC-FID analysis. Catalyst screening revealed that substituted acridines exhibited higher activity which also depended on the reaction type. For instance, in DT reaction di-*ortho*-phenyl substituted acridines (**2 a–d**, **2 p**, **2 q**, **2 t**) showed around twice as good the performance as its mono-*ortho*-phenyl substituted counterpart (**A-2**). The presence of a substituent in proximity to the nitrogen of the acridine (position 4) inhibited DT reaction (**2 i**, **2 k**, **2 n**, **2 r**). Surprisingly, in DA reaction, the 4-substituent considerably increased the productivity, presumably due to the alternative copper-involved path




**Scheme 5.** Evaluation of the catalytic activity.



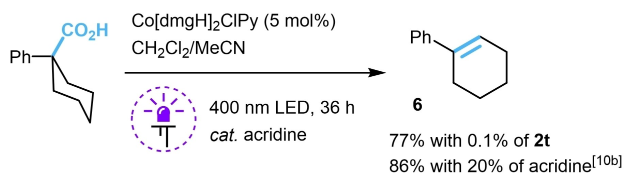

**Scheme 6.** Buried volume parameters according to the X-ray structural data. [a] Averaged values are given.

of catalyst breakdown. Overall, di-*tert*-butyl decorated acridine **2t** was found to be around three times more efficient catalyst in both DT and DA processes compared to the widespread catalyst **A-2**.

*Structural studies.* For nineteen acridines (**2a–f**, **h–p**, **s**, **t** and **A-1**, **A-2**) crystal structures were studied by X-ray diffraction analysis. The crystal structure of acridine **A-3** was reported previously.[11b] These data allowed us to evaluate the steric effect on the C-9 atom of acridines using the buried volume parameter $V_{Bur}$, which reflects the portion of a sphere buried by overlap with the substituent.[21] In Scheme 6, parameters averaged for the acridines of the similar *ortho*-substitution pattern are shown (the complete list of calculated parameters is given in Supporting Information). The smallest $V_{Bur}$ value of 55.3% was found for *ortho*-unsubstituted acridine **A-1**, while the largest value of 75.4% was for pentaphenylated compound **A-3**. Thus, increasing the *ortho*-substitution leads to increased $V_{Bur}$ value. Generally, catalysts with higher $V_{Bur}$ value work well in decarboxylative reactions. However, the best catalytic performance of acridine **2t** bearing bulky *tert*-butyl groups on the acridine core may reflect the need to protect benzene rings. Additionally, we tested the performance of acridine **2t** in cobalt-catalyzed dehydrodecarboxylation.[10b] Thus, 1-phenylcyclohexane-1-carboxylic acid gave alkene **6** in 77% yield with only 0.1% of **2t**. For comparison, 86% yield of **6** was achieved in the original work with 20% of unsubstituted acridine (Scheme 7).

## Conclusion

In summary, a practical approach to sterically hindered acridine photocatalysts via Negishi cross-coupling is described. The method utilizes readily available starting materials and a cheap nickel catalyst affording diverse acridines in a convenient and scalable protocol. The presence of bulky ortho-groups is believed to decrease the rate of catalyst degradation, which occurs via interception of the acridine-derived radical intermediate. As far as the efficiency of light absorbing catalysts is influenced by multiple factors, the accessibility of a wide pallet of acridines would facilitate the development of novel reactions in the field of acridine photocatalysis.




**Scheme 7.** Cobalt-catalyzed dehydrodecarboxylation.

## Experimental Section

### Substituted Acridine Preparation (General Procedure)

Corresponding 9-chloroacridine (0.5 mmol) and $[PPh_3]_2NiCl_2$ (65.4 mg, 0.1 mmol) were loaded into a Schlenk tube equipped with a stir bar. The tube was evacuated and filled with argon (three times). A solution of arylzinc chloride (3.8 ml, 0.4 M in THF, 1.5 mmol) was added [for the synthesis of **2p**, **s** (2,4,6-triisopropylphenyl)magnesium chloride was used], and the reaction mixture was stirred in an oil bath for 12 h at 60 °C. After cooling to room temperature, the mixture was quenched with saturated aqueous solution of $NH_4Cl$ (10 mL), treated with ethane-1,2-diamine (200 μL) and stirred for 30 min. Then, the mixture was extracted with $CH_2Cl_2$ (3×5 mL), the combined organic phases were dried over $Na_2SO_4$ and concentrated under reduced pressure. The residue was purified by column chromatography on silica gel.

CCDC 2167854–2167857, 2167864–2167875, 2167877, 2167878, 2168684, 2168685 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.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## References

[1] a) A. Schmidt, M. Liu, in: *Adv. Heterocycl. Chem., Vol. 115* (Eds.: E. F. V. Scriven, C. A. Ramsden), Academic Press, **2015**, pp. 287–353; b) J. Chiron, J.-P. Galy, *Synthesis* **2004**, *2004*, 313–325.

[2] a) G. Cholewiński, K. Dzierzbicka, A. M. Kołodziejczyk, *Pharmacol. Rep.* **2011**, *63*, 305–336; b) S. M. Sondhi, N. Singh, A. M. Lahoti, K. Bajaj, A. Kumar, O. Lozach, L. Meijer, *Bioorg. Med. Chem.* **2005**, *13*, 4291–4299.

[3] a) Á. Golcs, B. Á. Ádám, P. Vezse, P. Huszthy, T. Tóth, *Eur. J. Org. Chem.* **2021**, *2021*, 2485–2497; b) D. Zhang, X. Jiang, H. Yang, A. Martinez, M. Feng, Z. Dong, G. Gao, *Org. Biomol. Chem.* **2013**, *11*, 3375–3381; c) C. D. Geddes, *Dyes Pigm.* **2000**, *45*, 243–251.

[4] a) C. K. Prier, D. A. Rankic, D. W. C. MacMillan, *Chem. Rev.* **2013**, *113*, 5322–5363; b) J. W. Tucker, C. R. J. Stephenson, *J. Org. Chem.* **2012**, *77*, 1617–1622; c) L. Marzo, S. K. Pagire, O. Reiser, B. König, *Angew. Chem. Int. Ed.* **2018**, *57*, 10034–10072; *Angew. Chem.* **2018**, *130*, 10188–10228.

[5] a) S. Fukuzumi, H. Kotani, K. Ohkubo, S. Ogo, N. V. Tkachenko, H. Lemmetyinen, *J. Am. Chem. Soc.* **2004**, *126*, 1600–1601; b) S. Fukuzumi, K. Ohkubo, *Chem. Sci.* **2013**, *4*, 561–574.

[6] For reviews, see: a) K. A. Margrey, D. A. Nicewicz, *Acc. Chem. Res.* **2016**, *49*, 1997–2006; b) A. Tlili, S. Lakhdar, *Angew. Chem. Int. Ed.* **2021**, *60*, 19526–19549; c) N. A. Romero, D. A. Nicewicz, *Chem. Rev.* **2016**, *116*, 10075–10166.

[7] a) D. J. Wilger, J.-M. M. Grandjean, T. R. Lammert, D. A. Nicewicz, *Nat. Chem.* **2014**, *6*, 720–726; b) A. J. Perkowski, D. A. Nicewicz, *J. Am. Chem. Soc.* **2013**, *135*, 10334–10337; c) L. Pitzer, F. Sandfort, F. Strieth-Kalthoff, F. Glorius, *Angew. Chem. Int. Ed.* **2018**, *57*, 16219–16223; *Angew. Chem.* **2018**, *130*, 16453–16457; d) L. Pitzer, F. Sandfort, F. Strieth-Kalthoff, F. Glorius, *J. Am. Chem. Soc.* **2017**, *139*, 13652–13655; e) N. A. Romero, D. A. Nicewicz, *J. Am. Chem. Soc.* **2014**, *136*, 17024–17035; f) N. A. Romero, K. A. Margrey, N. E. Tay, D. A. Nicewicz, *Science* **2015**, *349*, 1326; g) N. E. S. Tay, D. A. Nicewicz, *J. Am. Chem. Soc.* **2017**, *139*, 16100–16104; h) J. D. Griffin, M. A. Zeller, D. A. Nicewicz, *J. Am. Chem. Soc.* **2015**, *137*, 11340–11348; i) K. A. Margrey, W. L. Czaplyski, D. A. Nicewicz, E. J. Alexanian, *J. Am. Chem. Soc.* **2018**, *140*, 4213–4217; j) Y.-X. Cao, G. Zhu, Y. Li, N. Le Breton, C. Gourlaouen, S. Choua, J. Boixel, H.-P. Jacquot de Rouville, J.-F. Soulé, *J. Am. Chem. Soc.* **2022**, *144*, 5902–5909.

[8] D. R. Weinberg, C. J. Gagliardi, J. F. Hull, C. F. Murphy, C. A. Kent, B. C. Westlake, A. Paul, D. H. Ess, D. G. McCafferty, T. J. Meyer, *Chem. Rev.* **2012**, *112*, 4016–4093.

[9] a) K. Okada, K. Okubo, M. Oda, *Tetrahedron Lett.* **1989**, *30*, 6733–6736; b) K. Okada, K. Okubo, M. Oda, *Tetrahedron Lett.* **1992**, *33*, 83–84.

[10] a) S. Jin, G. C. Haug, V. T. Nguyen, C. Flores-Hansen, H. D. Arman, O. V. Larionov, *ACS Catal.* **2019**, *9*, 9764–9774; b) V. T. Nguyen, V. D. Nguyen, G. C. Haug, H. T. Dang, S. Jin, Z. Li, C. Flores-Hansen, B. S. Benavides, H. D. Arman, O. V. Larionov, *ACS Catal.* **2019**, *9*, 9485–9498; c) H. T. Dang, G. C. Haug, V. T. Nguyen, N. T. H. Vuong, V. D. Nguyen, H. D. Arman, O. V. Larionov, *ACS Catal.* **2020**, *10*, 11448–11457; d) V. T. Nguyen, G. C. Haug, V. D. Nguyen, N. T. H. Vuong, H. D. Arman, O. V. Larionov, *Chem. Sci.* **2021**, *12*, 6429–6436; e) V. T. Nguyen, V. D. Nguyen, G. C. Haug, N. T. H. Vuong, H. T. Dang, H. D. Arman, O. V. Larionov, *Angew. Chem. Int. Ed.* **2020**, *59*, 7921–7927; *Angew. Chem.* **2020**, *132*, 7995–8001; f) V. T. Nguyen, G. C. Haug, V. D. Nguyen, N. T. H. Vuong, G. B. Karki, H. D. Arman, O. V. Larionov, *Chem. Sci.* **2022**, *13*, 4170–4179.

[11] a) I. A. Dmitriev, V. V. Levin, A. D. Dilman, *Org. Lett.* **2021**, *23*, 8973–8977; b) M. O. Zubkov, M. D. Kosobokov, V. V. Levin, A. D. Dilman, *Org. Lett.* **2022**, *24*, 2354–2358.

[12] M. O. Zubkov, M. D. Kosobokov, V. V. Levin, V. A. Kokorekin, A. A. Korlyukov, J. Hu, A. D. Dilman, *Chem. Sci.* **2020**, *11*, 737–741.

[13] a) Y. Liang, X. Zhang, D. W. C. MacMillan, *Nature* **2018**, *559*, 83–88; b) Z. Zuo, H. Cong, W. Li, J. Choi, G. C. Fu, D. W. C. MacMillan, *J. Am. Chem. Soc.* **2016**, *138*, 1832–1835; c) A. Noble, S. J. McCarver, D. W. C. MacMillan, *J. Am. Chem. Soc.* **2015**, *137*, 624–627.

[14] a) A. Bernthsen, *Liebigs Ann. Chem.* **1884**, *224*, 1–56; b) H. Koshima, K. Kutsunai, *Heterocycles* **2002**, *57*, 1299–1302.

[15] X. Ye, B. Xu, J. Sun, L. Dai, Y. Shao, Y. Zhang, J. Chen, *J. Org. Chem.* **2020**, *85*, 13004–13014.

[16] Z. Cao, Y. Zhu, X. Li, Y. He, J. Zhang, L. Xu, Y. Wei, *J. Org. Chem.* **2020**, *85*, 10167–10174.

[17] A. C. Benniston, A. Harriman, P. Li, J. P. Rostron, H. J. van Ramesdonk, M. M. Groeneveld, H. Zhang, J. W. Verhoeven, *J. Am. Chem. Soc.* **2005**, *127*, 16054–16064.

[18] a) D. Song, N. Zhang, P. Zhang, N. Zhang, W. Chen, L. Zhang, T. Guo, X. Gu, S. Ma, *Eur. J. Med. Chem.* **2021**, *221*, 113480; b) Q. Deng, Y. Shen, H. Zhu, T. Tu, *Chem. Commun.* **2017**, *53*, 13063–13066; c) Q. Zhao, C. Li, C. H. Senanayake, W. Tang, *Chem. Eur. J.* **2013**, *19*, 2261–2265.

[19] V. B. Phapale, D. J. Cárdenas, *Chem. Soc. Rev.* **2009**, *38*, 1598–1607.

[20] S. Kozuch, J. M. L. Martin, *ACS Catal.* **2012**, *2*, 2787–2794.

[21] a) L. Falivene, R. Credendino, A. Poater, A. Petta, L. Serra, R. Oliva, V. Scarano, L. Cavallo, *Organometallics* **2016**, *35*, 2286–2293; b) A. Poater, B. Cosenza, A. Correa, S. Giudice, F. Ragone, V. Scarano, L. Cavallo, *Eur. J. Inorg. Chem.* **2009**, *2009*, 1759–1766.